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Marks:73

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool:	Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: **Total Marks: 10**

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

```
A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
```

c. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>

Snippet 2: Form Submission Methods

A. GET
B. POST
C. PUT
D. DELETE

Correct answer: **B. POST**

Snippet 3: Navigation Elements

A.
B. <nav>
C. <section>
D. <header>

Correct answer: **B. <nav>**

Snippet 4: XML Syntax

A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note>

Correct answer: **B. <note><to>User</to><from>Admin</from></note>**

Snippet 5: Web Communication Protocols

A. FTP
B. HTTP
C. SMTP
D. SNMP

Correct answer: **B. HTTP**

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
   Answer: `<video>` tag. It is used to embed video content in a web page.
2. Identify the **appropriate HTTP method for form submission** and explain why.
   Answer: **POST**. It is used to send data (form fields) securely to the server.
3. Choose the **proper semantic element** for **navigation** and justify its usage.
   Answer: '**<nav>**'. It is used to define a section of navigation links.
4. Identify the **correct XML syntax** and explain the error in other options.
   Answer: Option B. `<note><to>User</to><from>Admin</from></note> .
   It correctly uses opening and closing tags in proper order.
5. Choose the **correct protocol** for **web communication** and justify your choice.
   Answer: **HTTP**. It is used for transferring data on the web (web pages, APIs, etc.).

Code of Conduct:

I S.Naveen kumar-192425176 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.Naveen

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Question 1: Select the correct HTML tag for video embedding and justify your answer.

Answer: Option B – <video src="video.mp4" controls></video>

Justification:

- <video> is the standard HTML5 element for embedding video directly in a page. It plays the file in the browser without any plug-in such as Flash.
- The **src** attribute gives the file to play, and the **controls** attribute shows the built-in play, pause, volume, seek and full-screen buttons, so the visitor can control the promotional video.
- Text placed between the opening and closing tags is shown as fallback content in browsers that do not support the element.
- It works with the HTML5 attributes poster, autoplay, muted, loop and preload, and with <track> for captions, which supports accessibility.

Why the other options are wrong:

Option	Tag	Reason
A	<media>	Not a valid HTML element; the browser treats it as an unknown tag and shows no player.
C	<movie>	Not a valid HTML element; it is not defined in any HTML standard.
D	<vid>	Not a valid HTML element; it is only an invented abbreviation.

Recommended implementation:

```
<video width="640" height="360" controls poster="thumb.jpg" preload="metadata">
  <source src="video.mp4" type="video/mp4">
  <source src="video.webm" type="video/webm">
  Your browser does not support the video tag.
</video>
```

Providing MP4 and WebM sources improves compatibility, and the poster image and preload="metadata" make the homepage load faster.

Question 2: Identify the appropriate HTTP method for form submission and explain why.

Answer: Option B – POST

Explanation:

- POST sends the form data inside the **body** of the HTTP request, so it does not appear in the URL, browser history, bookmarks or server access logs. This is essential for a registration form that carries passwords and personal details.
- POST has no practical limit on data size, whereas GET is limited by URL length.
- POST is meant for submitting data that creates or changes something on the server, such as a new user account. It is not cached and is not repeated silently by the browser.
- POST supports file uploads through enctype="multipart/form-data".

POST hides the data from the URL but does not encrypt it, so the form must also be served over **HTTPS**, and the server should hash passwords and validate input.

Why the other methods are not suitable:

Option	Reason
A. GET	Puts the data in the URL (for example /register?user=ali&pass=1234), so credentials are visible and stored in history and logs. GET is meant only for retrieving data.
C. PUT	Replaces or updates an existing resource at a known URL and is used in REST APIs. HTML forms cannot submit with PUT, and it is not meant for creating a registration.
D. DELETE	Removes a resource. It is the opposite of what a registration form should do, and HTML forms do not support it.

Recommended implementation:

```
<form action="/register" method="post">
  <input type="text" name="username" required>
  <input type="email" name="email" required>
  <input type="password" name="password" minlength="8" required>
  <button type="submit">Register</button>
</form>
```

Question 3: Choose the proper semantic element for navigation and justify its usage.

Answer: Option B – <nav>

Justification:

- <nav> is the HTML5 semantic element that marks a block of **navigation links** such as the main menu, a table of contents or pagination. Its name states its purpose, so the code is self-describing.
- **Accessibility:** screen readers announce it as a "navigation" landmark, so users can jump straight to the menu or skip it.
- **SEO:** search engines understand which links form the site structure.

- **Maintainability:** developers and CSS can find and style the menu easily, and the page follows web standards.

Why the other options are not proper:

Option	Reason
A.	A generic container with no meaning. It can hold a menu, but it tells browsers and assistive technology nothing.
C. <section>	Groups related content under a heading (a thematic section), not a set of navigation links.
D. <header>	Holds introductory content such as a logo or title. It may contain a <nav>, but it is not the navigation element itself.

Recommended implementation:

```
<header>
  <nav aria-label="Main menu">
    <ul>
      <li><a href="index.html">Home</a></li>
      <li><a href="about.html">About</a></li>
      <li><a href="contact.html">Contact</a></li>
    </ul>
  </nav>
</header>
```

Question 4: Identify the correct XML syntax and explain the error in other options.

Answer: Option B – <note><to>User</to><from>Admin</from></note>

Explanation: Option B is **well-formed** XML. It has a single root element (<note>), every start tag has a matching end tag, and the elements are properly nested, with <to> and <from> opened and closed inside <note>. XML is strict and case-sensitive, and a document that breaks these rules is rejected by the parser.

Errors in the other options:

Option	Code	Error
A	<note><to>User</to><from>Admin</from>	The root element <note> is never closed; the end tag </note> is missing.
C	<note><to>User</from></note>	The start tag <to> is closed by </from>, which does not match, so the nesting is improper. There is also no opening <from> tag.
D	<note to="User"></note	The end tag is incomplete: the closing ">" is missing in </note. (The attribute syntax to="User" itself is valid.)

Rules of well-formed XML applied:

- Exactly one root element.
- Every start tag has a matching, correctly spelled end tag.
- Elements are properly nested (the last element opened is the first closed).
- Tag names are case-sensitive and attribute values are quoted.

A complete configuration file following these rules:

```
<?xml version="1.0" encoding="UTF-8"?>
<note>
  <to>User</to>
  <from>Admin</from>
</note>
```

Question 5: Choose the correct protocol for web communication and justify your choice.

Answer: Option B – HTTP (in practice HTTPS)

Justification:

- HTTP (HyperText Transfer Protocol) is the application-layer protocol designed for the Web. A browser (client) sends a request and the web server returns a response containing HTML, CSS, images, JSON and other content.
- It runs over TCP, by default on port 80, and carries the methods (GET, POST), status codes (200, 404, 500) and headers used to deliver web pages and submit forms.
- For a registration form and user data the application should use **HTTPS** (HTTP over TLS, port 443). It encrypts the traffic and protects the confidentiality and integrity of the data.

Why the other protocols are not suitable:

Option	Protocol	Purpose (why it does not fit)
A	FTP (File Transfer Protocol, ports 20/21)	Transfers files between hosts; it does not deliver web pages to browsers.
C	SMTP (Simple Mail Transfer Protocol, port 25/587)	Sends e-mail between mail servers; not used for web pages.
D	SNMP (Simple Network Management Protocol, port 161)	Monitors and manages network devices such as routers and switches.

Therefore the startup should deliver the web application over HTTP, secured as HTTPS.

Criteria	Weighttage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	18	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	17	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	15	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	3	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand